Year 10 Foundation

REVISION PACK

Algebra 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Q	Working	Answer
substitution		
1.1	**(a)** $3 \times 4 + (-1) = 11$ **(b)** $4^2 - 2 \times (-1) = 18$	**(a) 11** **(b) 18**
1.2	**(a)** $5 \times 2 + (-5) = 5$ **(b)** $2^2 - 1 \times (-5) = 9$	**(a) 5** **(b) 9**
1.3	**(a)** $2 \times 6 + (-2) = 10$ **(b)** $6^2 - 3 \times (-2) = 42$	**(a) 10** **(b) 42**
1.4	**(a)** $2 \times 5 + (-3) = 7$ **(b)** $5^2 - 4 \times (-3) = 37$	**(a) 7** **(b) 37**
1.5	**(a)** $5 \times 3 + (-4) = 11$ **(b)** $3^2 - 2 \times (-4) = 17$	**(a) 11** **(b) 17**
1.6	**(a)** $2 \times 7 + (-1) = 13$ **(b)** $7^2 - 5 \times (-1) = 54$	**(a) 13** **(b) 54**
simplifying		
1.7		**(a)** $3x + 7y$ **(b)** $18ab$
1.8		**(a)** $4p + 4q$ **(b)** $14cd$
1.9		**(a)** $3a + 5b$ **(b)** $20mn$
1.10		**(a)** $5x + 5y$ **(b)** $24pq$

1.11		**(a)** $2m + 3n$ **(b)** $14xy$
1.12		**(a)** $4c + 5d$ **(b)** $24ab$
expanding brackets		
1.13	**(a)** $4 \times 3x = 12x$ $4 \times (-5) = -20$ **(b)** $x \times x = x^2$ $x \times 6 = 6x$	**(a)** $12x - 20$ **(b)** $x^2 + 6x$
1.14	**(a)** $6 \times 2x = 12x$ $6 \times 1 = 6$ **(b)** $x \times x = x^2$ $x \times (-3) = -3x$	**(a)** $12x + 6$ **(b)** $x^2 - 3x$
1.15	**(a)** $2 \times 5x = 10x$ $2 \times (-4) = -8$ **(b)** $x \times x = x^2$ $x \times 9 = 9x$	**(a)** $10x - 8$ **(b)** $x^2 + 9x$
1.16	**(a)** $7 \times 2x = 14x$ $7 \times (-3) = -21$ **(b)** $x \times x = x^2$ $x \times 8 = 8x$	**(a)** $14x - 21$ **(b)** $x^2 + 8x$
1.17	**(a)** $5 \times 3x = 15x$ $5 \times 1 = 5$ **(b)** $x \times x = x^2$ $x \times (-2) = -2x$	**(a)** $15x + 5$ **(b)** $x^2 - 2x$
1.18	**(a)** $3 \times 4x = 12x$ $3 \times (-7) = -21$ **(b)** $x \times x = x^2$ $x \times 1 = 1x$	**(a)** $12x - 21$ **(b)** $x^2 + x$
multiplying the same letter		

1.19	$3 \times 5 = 15$ $m \times m = m^2$	$15m^2$
1.20	$4 \times 2 = 8$ $x \times x = x^2$	$8x^2$
1.21	$6 \times 1 = 6$ $y \times y = y^2$	$6y^2$
1.22	$5 \times 4 = 20$ $a \times a = a^2$	$20a^2$
1.23	$2 \times 9 = 18$ $k \times k = k^2$	$18k^2$
1.24	$7 \times 3 = 21$ $n \times n = n^2$	$21n^2$

Algebra 1 · Solving equations (Lesson 2.4)

Q	Working	Answer
two-step equations		
2.1	$5x = 35$ $x = 7$	$x = 7$
2.2	$2x = 16$ $x = 8$	$x = 8$
2.3	$6x = 36$ $x = 6$	$x = 6$
2.4	$4x = 28$ $x = 7$	$x = 7$
2.5	$7x = 42$ $x = 6$	$x = 6$
2.6	$3x = 21$ $x = 7$	$x = 7$
equations with brackets		

2.7	$5x - 15 = 20$ $5x = 35$ $x = 7$	$x = 7$
2.8	$2x + 12 = 22$ $2x = 10$ $x = 5$	$x = 5$
2.9	$4x - 4 = 28$ $4x = 32$ $x = 8$	$x = 8$
2.10	$3x + 15 = 27$ $3x = 12$ $x = 4$	$x = 4$
2.11	$6x - 12 = 30$ $6x = 42$ $x = 7$	$x = 7$
2.12	$2x - 14 = 10$ $2x = 24$ $x = 12$	$x = 12$
unknowns on both sides		
2.13	$3x - 2 = 13$ $3x = 15$ $x = 5$	$x = 5$
2.14	$5x + 1 = 26$ $5x = 25$ $x = 5$	$x = 5$
2.15	$5x - 7 = 8$ $5x = 15$ $x = 3$	$x = 3$
2.16	$4x + 4 = 24$ $4x = 20$ $x = 5$	$x = 5$

2.17	$4x - 5 = 11$ $4x = 16$ $x = 4$	$x = 4$
2.18	$3x + 3 = 18$ $3x = 15$ $x = 5$	$x = 5$
equations with x over a number		
2.19	$\frac{x}{4} = 5$ $x = 5 \times 4$ $x = 20$	$x = 20$
2.20	$\frac{x}{5} = 6$ $x = 6 \times 5$ $x = 30$	$x = 30$
2.21	$\frac{x}{2} = 7$ $x = 7 \times 2$ $x = 14$	$x = 14$
2.22	$\frac{x}{6} = 3$ $x = 3 \times 6$ $x = 18$	$x = 18$
2.23	$\frac{x}{3} = 5$ $x = 5 \times 3$ $x = 15$	$x = 15$
2.24	$\frac{x}{5} = 6$ $x = 6 \times 5$ $x = 30$	$x = 30$